DATA ANALYTICS PROJECT REPORT

Customer Behavior Analysis

Understanding customer types, spending patterns, payment choices and satisfaction

Project Details	Information
Project	Customer Behavior Analysis
Dataset	supermarket_sales_cleaned.csv
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn and Jupyter Notebook
Report Type	Academic and Business Analysis Report

Prepared for academic and professional portfolio use

EXECUTIVE SUMMARY

This project examines customer behaviour in supermarket transactions to understand how different customer groups spend, which payment methods they prefer and how customers evaluate their shopping experience. The supplied project summary reports an average customer rating of 6.96 and a standard deviation of 1.73. E-wallet is identified as the most preferred payment method, while overall satisfaction is described as generally positive. These findings can help a supermarket improve customer experience, payment convenience and retention. Exact group-level averages and counts should be inserted from the executed notebook before final submission.

1. INTRODUCTION

Customer behaviour analysis helps retailers understand how shoppers interact with products, payment systems and services. Transaction records provide evidence about spending patterns and customer feedback. By organizing these records into meaningful customer segments, a supermarket can identify opportunities to improve convenience, satisfaction and repeat visits. This project uses descriptive analytics to summarize customer-related patterns in the supplied cleaned dataset.

2. PROJECT OBJECTIVES

- Compare the spending behaviour of different customer types.
- Examine spending differences across gender groups.
- Identify the most frequently used payment method.
- Study the distribution of customer ratings.
- Assess the overall customer satisfaction pattern.
- Convert the analysis into practical recommendations for supermarket management.

3. DATASET DESCRIPTION

The cleaned supermarket dataset contains transaction-level information used to study customer type, gender, payment method and customer ratings. The supplied project specification states that the dataset contains 2,000 records and 16 features. The analysis focuses on customer-related fields and does not assume additional variables beyond those available in the file.

Attribute	Description
Dataset file	supermarket_sales_cleaned.csv
Records	2,000
Features	16
Analysis environment	Jupyter Notebook with Python libraries

4. DATA PREPARATION AND VALIDATION

Data preparation should begin by loading the CSV file into a Pandas DataFrame and inspecting its dimensions, column names, data types and sample records. Missing values, duplicate rows and inconsistent category labels should be checked. Numeric fields used for spending and ratings should be converted to suitable numeric data types. Categorical values such as customer type, gender and payment method should be standardized so that spelling or spacing differences do not create separate groups. Validation should include checking that ratings fall within the scale used by the dataset and that spending values are non-negative where appropriate. The project summary identifies the cleaned input file; therefore, any further cleaning should be documented rather than assumed.

5. EXPLORATORY AND STATISTICAL ANALYSIS

The analysis should use descriptive statistics and grouped summaries. For spending, calculate transaction count, total spending, mean spending and median spending by customer type and gender. For payment preference, calculate both transaction frequency and percentage share. For ratings, calculate mean, median, standard deviation, minimum, maximum and frequency by rating value. These measures answer different questions: totals indicate contribution to overall sales, while averages describe the typical transaction within a group. Where group sizes differ, percentages and per-transaction averages provide fairer comparisons.

6. VISUALIZATION AND INTERPRETATION

Visualization	Purpose and interpretation
Customer Type Spending	Compare total or average spending between customer types. Use the same spending measure for each group; a higher total may reflect more transactions rather than higher spending per customer.
Gender Spending	Compare average spending across gender categories. Interpret differences cautiously and avoid assuming that gender itself causes spending behaviour.
Customer Rating Distribution	Show how ratings are distributed across the available rating scale. The mean of 6.96 and standard deviation of 1.73 provide a numerical summary, while the chart reveals concentration and spread.
Payment Method Analysis	Compare transaction counts by payment method. The supplied finding identifies E-wallet as the leading preference; exact percentages should be taken from the notebook.
Customer Behavior Heatmap	Use a heatmap of appropriately aggregated customer measures to explore patterns between customer groups and behaviour. State the aggregation method and sample sizes in the chart notes.

7. KEY FINDINGS

- The reported average rating is 6.96.

- The reported standard deviation of customer rating is 1.73.
- E-wallet is reported as the most preferred payment method.
- Overall customer satisfaction is described as generally positive.

8. BUSINESS INSIGHTS

An average rating of 6.96 suggests a broadly favourable evaluation on the dataset's rating scale, but the interpretation depends on the scale maximum and rating question. The standard deviation of 1.73 indicates that individual ratings vary around the mean. E-wallet preference highlights the importance of convenient digital payment options. Customer-type and gender comparisons can support segmentation, but managers should examine average spending alongside transaction counts to avoid misleading conclusions.

9. BUSINESS RECOMMENDATIONS

- Maintain reliable and fast E-wallet payment facilities while continuing to support other payment options.
- Investigate lower-rating transactions and identify recurring service or product issues.
- Use customer-type segments to tailor loyalty offers, without excluding less frequent shoppers.
- Compare average spend, transaction frequency and total revenue together when evaluating segments.
- Collect feedback regularly and monitor rating trends over time.
- Test promotional campaigns and compare results against a baseline before expanding them.

10. LIMITATIONS AND FURTHER WORK

Only the overall rating mean, standard deviation, leading payment method and broad satisfaction description were supplied as numerical findings. Exact medians, group averages, payment shares, sample sizes and statistical significance were not provided. No causal conclusions can be drawn from descriptive comparisons alone. Further work should calculate group-level summaries, confidence intervals where appropriate and tests selected according to the data distribution and research question.

11. CONCLUSION

Customer behaviour analysis provides a structured view of spending, payment preference and satisfaction. The reported rating statistics and E-wallet preference offer useful starting points for management decisions. A final evidence-based report should supplement these summary findings with verified group-level values from the notebook and use the results to monitor customer experience continuously.

APPENDIX: REPRODUCIBILITY CHECKLIST

- Retain the original dataset and the cleaned dataset separately.
- Record the exact cleaning steps and the number of rows removed or modified.

- Run all notebook cells from top to bottom before exporting the final notebook.
- Label every chart with a clear title, axis names and units.
- Add exact calculated values to the report only after verifying them from the notebook output.
- Keep the analysis code, report and README together in the project repository.